

Andrea Zingarofalo

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📍 Brussels, Belgium

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Education

PhD	Université libre de Bruxelles , Quantum Information Theory	Brussels, Belgium
	- Thesis: Self-Testing and Device-Independent Verification of Quantum Computation - Advisor: Dr. S. Pironio	Sept 2025 – present
Diplôme d'Ingénieur	Advanced School ISUFI (Excellence Program) , Natural Sciences	Lecce, Italy
Li-cenza	70/70 cum laude — ~10% acceptance rate - Modules: Group Theory and Symmetries, Celestial Mechanics, Astroparticle Physics, Applied Physics I-II, Philosophy of Science, ICT for Research, Public Speaking, Scientific Writing	Sept 2020 – July 2025
Exchange Program	École normale supérieure , Theoretical Physics – Master ICFP M2	Paris, France
	Modules: General Relativity, Quantum Field Theory, Statistical Field Theory, Algorithms & Computations	Sept 2024 – July 2025
MSc	University of Salento , Theoretical Physics	Lecce, Italy
	110/110 cum laude, GPA: 29.8/30 - Modules: Quantum Field Theory I-II, QCD, Statistical Mechanics, Quantum Computing, Physics of Complex Systems, Data Analysis, Mathematical Methods for AI - Thesis: Exploring the boundaries of quantum correlations: Bell inequalities, convex duality, and self-testing - Advisors: Prof. L. Martina & Prof. J.-D. Bancal	Sept 2023 – July 2025
BSc	University of Salento , Physics	Lecce, Italy
	110/110 cum laude, GPA: 30/30 - Electives: Nuclear Physics, Astronomy, Foundations of Physics - Thesis: Study of one-dimensional quantum integrable systems via the Bethe Ansatz - Advisor: Prof. L. Martina	Sept 2020 – July 2023

Experience

Laboratoire d'Information Quantique – Université libre de Bruxelles , Doctoral Researcher	Brussels, Belgium
- Research on self-testing and device-independent protocols for certifying components of quantum computation, from multipartite entanglement to error-correcting codes - Supervisor: Dr. S. Pironio	Sept 2025 – present 8 months
Institut de Physique Théorique – CEA Paris-Saclay , Visiting PhD Student	Paris, France
- Research on verification protocols for quantum computation - Supervisor: Prof. J.-D. Bancal	Oct 2025 – Feb 2026 5 months
Mindrift , AI Model Training Specialist (Physics)	Remote
- Contributed to training and evaluation of AI models by providing expert-level physics supervision and structured feedback - Analyzed model outputs to identify systematic failure modes and guide iterative model refinement	Nov 2025 – present 6 months

Institut de Physique Théorique – CEA Paris-Saclay , Research Intern - Derived analytically new Bell inequalities for partially entangled states - Implemented SDP relaxations to numerically estimate quantum bounds of Bell inequalities via the Navascués-Pironio-Acín hierarchy - Supervisor: Prof. J.-D. Bancal	Paris, France Mar 2025 – July 2025 5 months
Mathematical Physics Group – University of Salento (with University of Glasgow) , Research Intern - Derived multi-kink solutions of a generalized Fokker-Planck equation via Hirota's bilinear method - Developed automated Mathematica scripts for symbolic generation of soliton solutions - Supervisor: Prof. L. Martina	Lecce, Italy Mar 2024 – July 2024 5 months
Self-employed , Private Tutor in Mathematics, Physics, and Chemistry High-school and university students	Italy 2019 – present 7 years

Publications

Integrability properties and multi-kink solutions of a generalised Fokker-Planck equation F. Giglio, G. Landolfi, L. Martina, A. Zingarofalo (Journal of Physics A: Mathematical and Theoretical, 58(16):165202)	Apr 2025
Quantum correlations: A geometric perspective A. Zingarofalo (Preprint – to be published)	2025
Il conflitto tra meccanica quantistica e senso comune A. Zingarofalo (Il Chiasmo by Treccani – The Italian Encyclopaedia Institute)	2024
Il ruolo delle simmetrie nel problema di Keplero A. Zingarofalo (Rete di Idee, Udine 2022 – Forum Editrice, pages 81–100)	2023

Awards and Scholarships

Member of Nova – The Global Top Talent Network (top 3% worldwide) Recognizing the top 3% of professionals worldwide across STEM, business, and innovation	2025
Winner of RIASISSU "Rete di Idee", XIII edition	2025
Top 100 STEM Graduates in Italy (Class of 2025) Selected by Almalaurea, the national consortium of Italian universities	2025
ISUFI "Research & Innovation" Grant for International Research Mobility	2025
Cité Internationale Universitaire de Paris Scholarship Accommodation scholarship	2024
Winner of RIASISSU "Rete di Idee", X edition	2022
ISUFI Scholarship at the University of Salento Awarded for academic excellence (full accommodation and a stipend)	2020

Senior Finalist at XVI Italian Astronomy Olympiads

2018

- Ranked among the top 30 out of over 6000 participants nationwide

Skills

Programming: Python (NumPy, SciPy, Matplotlib, pandas), C/C++, Julia, Bash, Javascript, Mathematica, Matlab

Quantum Libraries: Qiskit, PennyLane, CVXPY, MOSEK, ncpol2sdpa

Data Analysis: ROOT framework, scikit-learn, TensorFlow

Workflow: LaTeX, Markdown, git, Inkscape, Microsoft Office, UNIX/Linux

Languages

Italian: Native

English: C1

French: B2

German: A2